In exercise problems 7 – 15, use the graph to find the limits (if it exists). If the limit does not exist, explain why?

(7) $\lim_{x\to 3}(4-x)$.

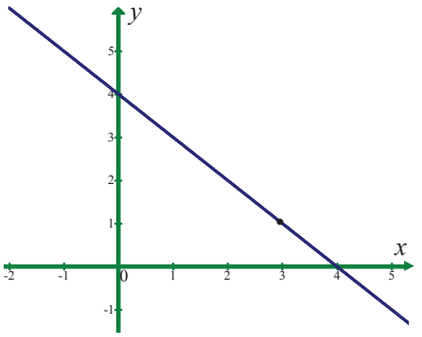


Fig. 9.13

(8) $\lim_{x\to 1}(x^2+2)$.

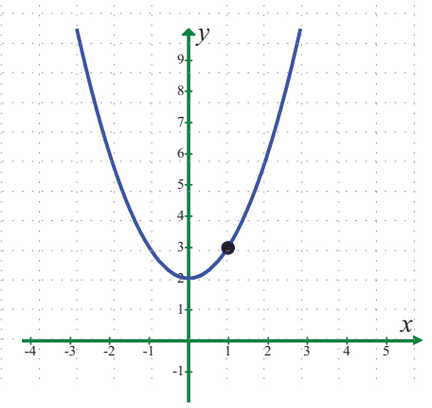


Fig. 9.14

(9) $\lim_{x\to 2} f(x)$

where $f(x) = \begin{cases} 4-x, & x \neq 2 \\ 0, & x = 2 \end{cases}$.

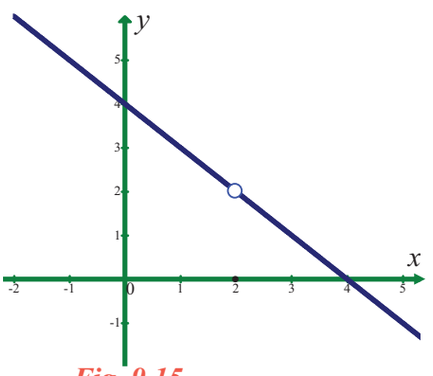


Fig. 9.15

(10) $\lim_{x\to 1} f(x)$

where $f(x) = \begin{cases} x^2+2, & x \neq 1 \\ 1, & x = 1 \end{cases}$.

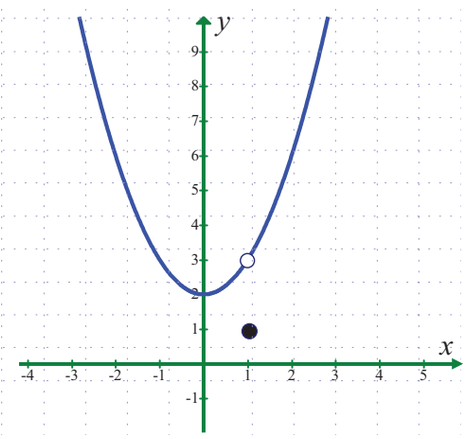


Fig. 9.16

(11) $\lim_{x\to 3} \dfrac{1}{x-3}$

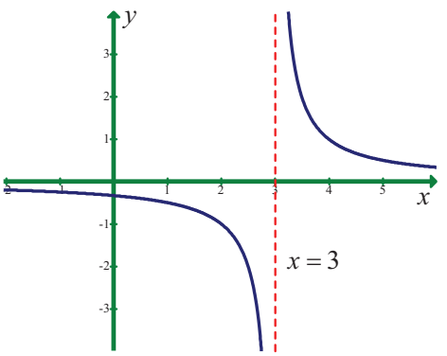


Fig. 9.17

(12) $\lim_{x\to 5} \dfrac{|x-5|}{x-5}$

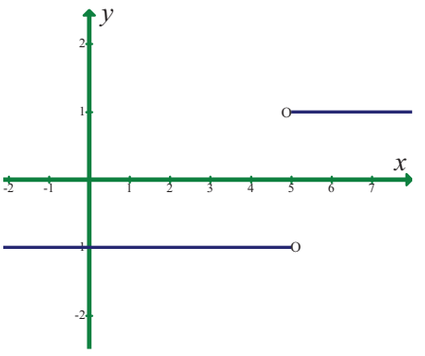


Fig. 9.18

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